Experiment 16: Time Series Analysis of Website Traffic

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose


data = pd.read_csv('website_traffic.csv', parse_dates=['Date'],
index_col='Date')


decomposition = seasonal_decompose(data['page_views'],
model='additive')

decomposition.plot()

plt.show()


data['page_views_Z'] = (data['page_views'] -
data['page_views'].mean()) / data['page_views'].std()


anomalies = data[abs(data['page_views_Z']) > 3]

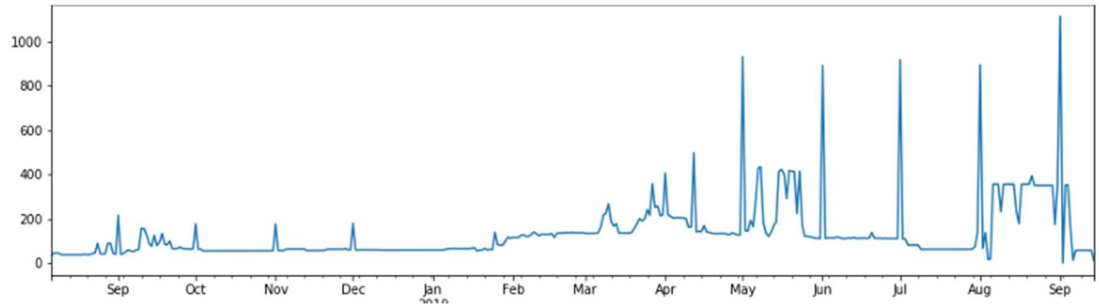

plt.plot(data.index, data['page_views'], label='page_views')

plt.scatter(anomalies.index, anomalies['page_views'], color='red',
label='Anomalies')

plt.legend()

plt.show()
```

Output:



Result:

Thus , Time Series Analysis of Website Traffic code was analysed and executed successfully.